

Demo: Build a Speaker

Wind, align, measure, and prove a paper-cup loudspeaker

Lesson Demo D · 45–60 minutes · after Lesson 4 — isolated battery audio only

Boundary before the build

Use only a battery-powered, current-limited audio amplifier, initially at minimum level. The coil connects through a 100 Ω , 1 W resistor. An adult secures the magnet and controls power. Disconnect for every resistance, alignment, or wiring check. Stop for heat, odor, pain, damaged insulation, distortion, or a magnet that can move.

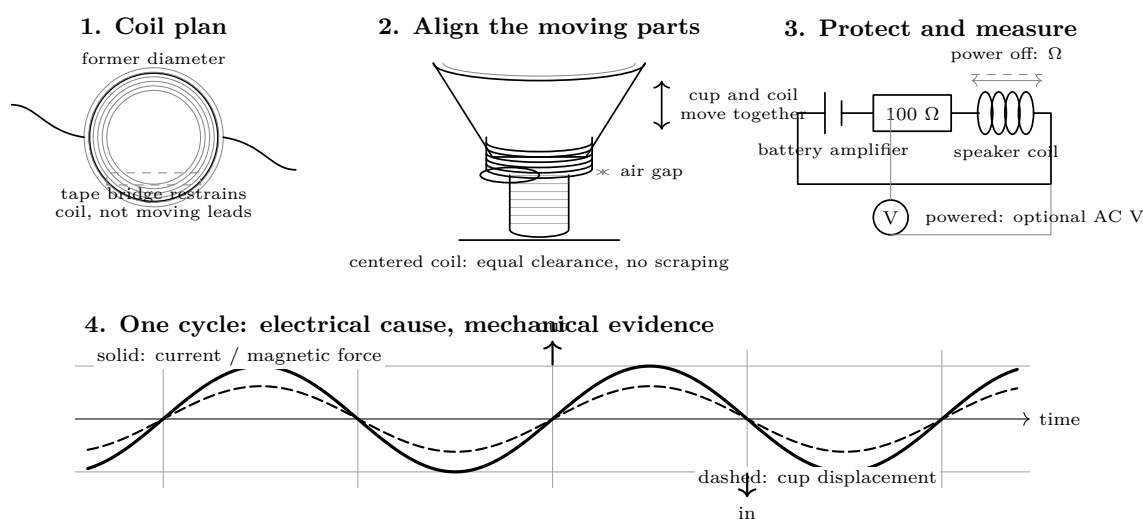
What you need

- 30–32 AWG enamelled wire; card former; sandpaper
- Paper cup, low-tack tape, narrow paper motion flag
- Ceramic magnet, or neodymium magnet secured by the adult
- Battery audio amplifier and tone source; 100 Ω , 1 W resistor
- Digital multimeter; ruler; optional spectrum-display app

Predict first

Draw the cup position for positive, zero, and negative coil current. Will a steady current make a continuing note? Which check should be made before power: sound, resistance, or frequency? Predict where an added paper clip will move the cup's strongest response.

Assembly atlas



Qualify the build in stages

A. Electrical check — power disconnected

1. Scrape both tails to bright copper. Measure the meter leads touching each other, then measure across the coil. Record R_{leads} and R_{measured} ; estimate $R_{\text{coil}} = R_{\text{measured}} - R_{\text{leads}}$.

- Flex each tail gently while watching the display. A stable finite reading is the pass condition. Open circuit means enamel or a break; a near-zero result means verify the range and look for a bypass.

B. Mechanical check — still unpowered

- Tape the coil flat and concentric to the cup. Adult secures the magnet. Move the cup lightly through its expected travel. Listen and feel: silence, free motion, and equal clearance pass; scrape or sticking fails.
- Attach a tiny paper flag to the rim. Confirm the 100 Ω resistor, insulation, and minimum volume before the adult permits power.

C. Motion and response

- Play 25 Hz briefly. Raise level only until the flag's alternating motion is visible. Ask: why can motion be visible before a clear pitch is heard?
- Without changing the level, try 100, 200, 400, 800, and 1200 Hz. Mark sound as 0 (none), 1 (faint), 2 (clear), or 3 (strong). If the meter is rated for the frequency, record AC volts across the coil; otherwise write "bandwidth not established."

Hz	V_{coil}	Level 0–3	Motion	Buzz / scrape / other evidence
25	_____	_____	_____	_____
100	_____	_____	_____	_____
200	_____	_____	_____	_____
400	_____	_____	_____	_____
800	_____	_____	_____	_____
1200	_____	_____	_____	_____

Diagnose; do not guess

What you know	What it suggests	Next isolated check
Meter reads open	Enamel remains or wire is broken	Re-scrape; inspect and flex one tail at a time.
Resistance passes; no motion	Signal path or joint has failed	Test the amplifier with an approved load; trace each connection.
Cup moves; sound is faint	Cone coupling or gap is poor	Power off; reduce loose tape and gap without permitting contact.
Scratch once per cycle	Coil is off-center	Re-center until unpowered travel is silent.
One frequency buzzes	Loose material or a cup resonance	Repeat just above and below; press suspect tape only while unpowered.
Warmth, odor, harsh distortion	Excess drive or damaged coil	Disconnect immediately; cool, inspect, and remeasure before reuse.

Final proof

The speaker is demonstrated only when every claim has its own evidence:

- **Electrical:** coil resistance is finite, repeatable, and stable under a gentle tail-flex test.
- **Mechanical:** the unpowered cup travels freely and silently.
- **Transduction:** 25 Hz produces visible alternating motion.
- **Acoustic:** at least two commanded frequencies produce repeatable, distinguishable sound without a stop condition.

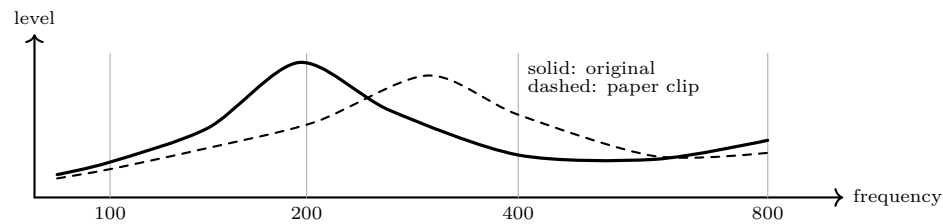
Explain only what the evidence earns

Alternating current in the permanent magnet's field gives the coil an alternating force. The cup follows and makes alternating pressure in the air. Steady DC gives at most a one-time displacement plus heating, not a continuous tone.

Response is not flat: electrical impedance, moving mass, stiffness, magnetic clearance, cup resonances, and hearing all affect observed level. Resistance proves continuity; motion proves force reached the cup; repeatable sound proves air was driven. No one observation proves the whole chain.

Controlled change: add moving mass

Tape one paper clip near the rim. Predict, then repeat 100, 200, 400, and 800 Hz at the same source setting. Do not compare trials at different volume.

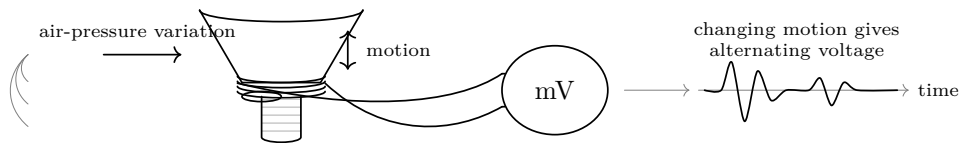


Hz	Original 0–3	With mass 0–3	Change heard or seen
100	_____	_____	_____
200	_____	_____	_____
400	_____	_____	_____
800	_____	_____	_____

Read your data. Which frequency was strongest before and after? Did mass move the response as predicted? State one conclusion and one uncertainty.

Reverse the chain: cup motion to voltage

Disconnect the amplifier and resistor. On the meter’s lowest AC-voltage range, measure directly across the coil while speaking near the cup, then gently tapping the table beside it. Do not strike the magnet or cup. Motion now moves the coil through the field and develops a voltage.



Input	Peak AC mV	Repeatable?	What moved the cup?
Quiet room	_____	_____	_____
Voice near cup	_____	_____	_____
Tap beside base	_____	_____	_____

Close the argument. Reverse the electrical, mechanical, and acoustic labels for microphone mode. What shows reciprocity? Why can near-zero fail to prove that no voltage exists?

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